# Aadit Kshirsagar

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## EDUCATION

**University of Southern California** — Aug 2025 - May 2027
*Master of Science in Computer Science* — *GPA:3.5/4*

**Savitribai Phule Pune University** — Aug 2020 - Jun 2024
*Bachelor of Engineering in Computer Science* — *GPA:3.61/4*

## WORK EXPERIENCE

**Tredence Inc** — **Jun 2026 – Aug 2026**
*AI Intern* — *San Jose, CA*

- Built an **LLMs-powered platform** to optimize the Statement of Work and Purchase Order lifecycle, integrating **automated data ingestion, forecasting and analytical workflows** to improve visibility across customer engagements.
- Developed and deployed an **analytics agent for natural-language querying** of operational data and forecasting outcomes, enabling 150+ internal users to extract actionable insights for delivery and planning decisions.
- Designed a **LangGraph-based agentic architecture** to detect commercial order status and analyze delivery progress, enabling automated monitoring and decision support through an end-to-end proof of concept.

**USC Viterbi School of Engineering** — **Jan 2026 - May 2026**
*Research Assistant* — *Los Angeles, CA*

- Built an **Alzheimer's disease classification pipeline in PyTorch** using OASIS brain MRI data, incorporating ResNet-18 fine-tuning and multi-class evaluation across four dementia stages.
- Engineered a training pipeline with **patient-level stratified splits, focal loss, class-balanced sampling** and MRI-specific augmentations to mitigate class imbalance and data leakage, **achieving 87% classification accuracy**.

**Calsoft Pvt Ltd** — **Apr 2024 - Jun 2025**
*Data Science Intern* — *Pune, India*

- Led the **end-to-end development of enterprise GenAI systems**, from system design and implementation to optimization and production deployment.
- Built a **multi-pipeline, agent-based RAG system** that improved test-case retrieval relevance to **95%** while enabling semantic search across legacy test assets.
- Deployed an **on-premises AI solution using open-source LLMs**, reducing manual test-case generation by **80%** and accelerating QA workflows.

**Intellect Design Arena Ltd** — **Sep 2023 - Mar 2024**
*AI/ML Intern* — *Pune, India*

- Engineered an **intelligent banking agent** for iGTB Copilot to analyze **customer payment patterns and transaction data**, enabling Relationship Managers to identify trends and derive actionable insights for client engagement.
- Integrated **LLMs with customer payment data** and developed **prompt-driven workflows** for a banking Copilot proof of concept, automating **over 70%** of the payment onboarding process**.**

## PROJECTS

**NextVisit** — next-visit-six.vercel.app/

- Built and deployed an **end-to-end restaurant hygiene intelligence platform** using Python, FastAPI and TypeScript, integrating public health inspection data across 7 jurisdictions to deliver predictive risk insights, inspection histories and location-based restaurant comparisons.
- Developed **city-specific machine learning models using LightGBM and scikit-learn**, incorporating historical inspection outcomes, temporal features and violation patterns to predict future inspection risks, achieving **0.949 PR-AUC** on held-out data.
- Engineered **data ingestion and feature engineering pipelines** with temporal validation, baseline benchmarking, calibration analysis and feature ablation studies to ensure reproducible model evaluation across heterogeneous inspection datasets.

**Agent Gary - *Origin Weekend x Replit Finalist*** — Watch on YouTube

- Led a four-person team that built Gary, a **Twilio and Deepgram voice agent on FastAPI and MCP** that runs from a phone call with no app, so older adults can pay bills, order food and book rides and services while family stays in the loop.
- Designed and implemented **payment and inbox pipelines** with policy checks, spoken read-back, single-use transaction confirmation and Gmail/Calendar ingestion for classifying bills, appointments and suspected scams.

## TECHNICAL SKILLS

- **Programming**: Python, Java, SQL, JavaScript
- **AI/ML & GenAI**: AI Agents, LangChain, LangGraph, PyTorch, LLMs, RAG, Embeddings, Vector Databases, MCP, Hugging Face, OpenCV, MLOps, LLMOps, TensorFlow, NumPy
- **Cloud & DevOps**: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Git
- **Web and Backend:** React, Next.js, Node.js, Django, Flask, FastAPI, Streamlit
- **Databases and Data Engineering:** MongoDB, PostgreSQL, MySQL, SQLite, PySpark